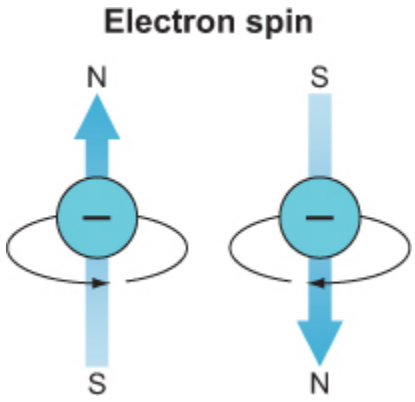
9.3. The 1D Ising model

In this section we're going to apply MARL to solve a real physics problem that was first described in the early 1920s by physicist Wilhelm Lenz and his student Ernst Ising. But first, a brief physics lesson. Physicists were trying to understand the behavior of magnetic materials such as iron by mathematical models. A piece of iron that you can hold in your hand is a collection of iron atoms that are grouped together by metallic bonding. An atom is composed of a nucleus of protons (positively charged), neutrons (no charge), and an outer "shell" of electrons (negatively charged). Electrons, like other elementary particles, have a property called *spin*, which is quantized such that an electron can only have a spin-up or spin-down at any time (figure 9.8).

Figure 9.8. Electrons are negatively charged elementary particles that surround the nucleus of every atom. They have a property called spin, and it can either be spin-up or spin-down. Since they are charged particles, they generate a magnetic field, and the direction they spin determines the orientation of the poles (north or south) of the magnetic field.



The spin property can be thought of as the electron rotating either clockwise or counter-clockwise; this is not literally true, but it suffices for our purposes. When a charged object rotates, it creates a magnetic field, so if you took a rubber balloon, gave it a static charge by rubbing it on the carpet, and then spun it around, you would have yourself a balloon magnet (albeit an extremely weak magnet). Electrons likewise create a magnetic field by virtue of their spin and electric charge, so electrons really are very tiny magnets, and since all iron atoms have electrons, the entire piece of